

Self-Assessment Evidence

Results Screen

1. Number theory - SIT281 - Crypt...

Module assessment - SIT281 - Crypt...

number theory part 1-exam

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OnTrack

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number theory part 1-exam

Click on a question number to see how your answers were marked and, where available, full solutions.

Question Number	Score	
Question 1	3 / 3	Review
Question 2	3 / 3	Review
Question 3	1 / 1	Review
Question 4	1 / 1	Review
Question 5	1 / 1	Review
Question 6	1 / 1	Review
Question 7	1 / 1	Review
Question 8	1 / 1	Review
Total	12 / 12 (100%)	

Performance Summary

Exam Name:	number theory part 1-exam
Session ID:	03270705203
Exam Start:	Sun Jul 12 2026 20:45:59
Exam Stop:	Sun Jul 12 2026 21:52:36
Time Spent:	1:06:36

Print this results summary

Working

$$\begin{array}{l|l|l} \textcircled{1} \gcd(88, 67) \Rightarrow & \underline{88} & \underline{67} \\ 88 - 67 \times 1 = 21 & 1 - 0 = 1 & 0 - 1 \times 1 = -1 \\ 67 - 21 \times 3 = 4 & 0 - 1 \times 3 = -3 & 1 - (-1) \cdot 3 = 4 \\ 21 - 4 \times 5 = 1 & 1 - -3 \times 5 = 16 & -1 - (4 \times 5) = -21 \end{array}$$
$$1 = 88 \times \underline{16} + 67 \times \underline{-21}$$
$$1 = 88 \times 16 - 67 \times 21$$

$$\textcircled{2} \text{ gcd}(47, 87) \Rightarrow$$

$$87 - 47 \times 1 = 40$$

$$47 - 40 \times 1 = 7$$

$$40 - 7 \times 5 = 5$$

$$7 - 5 \times 1 = 2$$

$$5 - 2 \times 2 = 1 //$$

$$\begin{array}{c|l} \underline{47} & \\ \hline 0 - 1 = -1 & \\ 1 - -1 = 2 & \\ -1 - 2 \times 5 = -11 & \\ 2 - -11 \times 1 = 13 & \\ -11 - 13 \times 2 = -37 & \end{array}$$

$$\begin{array}{c|l} \underline{87} & \\ \hline 1 - 0 = 1 & \\ 0 - 1 \times 1 = -1 & \\ 1 - (-1) \cdot 5 = 6 & \\ -1 - 6 \times 1 = -7 & \\ 6 - -7 \times 2 = 20 & \end{array}$$

$$1 \equiv 47 \times \underline{-37} + 87 \times \underline{20}$$

$$\textcircled{3} \text{ modular inverse of } 89 \text{ mod } 28$$

$$89 - 28 \times 3 = 5 \Rightarrow 89 = 28 \times 3 + 5$$

$$\text{gcd}(89, 28) \Rightarrow$$

$$89 - 28 \times 3 = 5$$

$$28 - 5 \times 5 = 3$$

$$5 - 3 \times 1 = 2$$

$$3 - 2 \times 1 = 1$$

$$\begin{array}{c|l} \underline{89} & \\ \hline 1 - 0 = 1 & \\ 0 - 1 \times 5 = -5 & \\ 1 - -5 \times 1 = 6 & \\ -5 - 6 \times 1 = \underline{-11} & \swarrow \text{inverse} \end{array}$$

$$-11 + 28 = +17$$

$$\therefore \Rightarrow 17 \text{ mod } 28 //$$

④ inverse of 79 mod 91

$$\gcd(79, 91) \Rightarrow$$

$$91 - 79 \times 1 = 12$$

$$79 - 12 \times 6 = 7$$

$$12 - 7 \times 1 = 5$$

$$7 - 5 \times 1 = 2$$

$$5 - 2 \times 2 = 1$$

$$\underline{\underline{79}}$$

$$0 - 1 \times 1 = -1$$

$$1 - -1 \times 6 = 7$$

$$-1 - 7 = -8$$

$$7 - -8 = 15$$

$$-8 - 15 \times 2 = -\underline{\underline{38}}$$

inverse

$$-38 + 91 = 53$$

$$\therefore \Rightarrow 53 \text{ mod } 91$$

$$\textcircled{5} \quad n = \overset{a}{\underline{0}} \text{ mod } \overset{m}{\underline{5}}$$

$$n = \overset{b}{\underline{10}} \text{ mod } \overset{m}{\underline{19}}$$

$$\gcd(5, 19) \Rightarrow$$

$$19 - 5 \times 3 = 4$$

$$5 - 4 \times 1 = 1$$

$$\underline{\underline{5}}$$

$$-3$$

$$1 - -3 = \underline{\underline{\frac{4}{5}}}$$

$$\underline{\underline{19}}$$

$$1$$

$$-1$$

$$\underline{\underline{6}}$$

$$n \equiv bms + ant \text{ (mod } m \cdot n)$$

$$n \equiv 10 \times 5 \times 4 + 0 \text{ (mod } 5 \times 19)$$

$$n \equiv 200 \text{ (mod } 95)$$

$$n \equiv 10 \text{ mod } 95$$

$$\left| \begin{array}{l} 200 = 95 \times 2 + 10 \end{array} \right.$$

$$\textcircled{6} \quad n \equiv 0 \pmod{2} \\ n \equiv 26 \pmod{29}$$

$$\gcd(2, 29) = 1 \quad \begin{array}{c|c} 2 & 29 \\ \hline 29 - 2 \times 14 = 1 & -14 \quad 1 \end{array}$$

$$\begin{aligned} n &\equiv bms + cut \pmod{m \times n} \\ &\equiv 26 \times 2 \times -14 + 0 \pmod{2 \times 29} \\ &\equiv -728 \pmod{58} \\ &\equiv 26 \pmod{58} \end{aligned} \quad \left| \begin{array}{l} -728 + 58 \times 13 = 26 \end{array} \right.$$

$$\textcircled{7} \quad 70n \equiv 41 \pmod{61} \quad (0 \leq n \leq 60)$$

$$70 \Rightarrow 61 \times 1 + 9$$

$$9n \equiv 41 \pmod{61}$$

$$\text{inverse} \Rightarrow (9 \pmod{61})$$

$$\gcd(9, 61) = 1 \quad \begin{array}{c|c} 9 & 61 \\ \hline 61 - 9 \times 6 = 7 & -6 \\ 9 - 7 \times 1 = 2 & 1 + 6 = 7 \\ 7 - 2 \times 3 = 1 & -6 - 7 \times 3 = -27 \end{array}$$

$$9x - 27 \equiv 1 \pmod{61}$$

$$-27 \equiv 34 \pmod{61}$$

$$61 - 27 = 34$$

$$\text{back to } 9n \equiv 41 \pmod{61} \Rightarrow$$

$$\times 34 \Rightarrow n \equiv 41 \times 34 \pmod{61}$$

$$n \equiv 1394 \pmod{61}$$

$$n = 52$$

$$1394 - 61 \times 22 = 52$$

let's verify $\Rightarrow$

$$70n \equiv 41 \pmod{61}$$

$$70 \times 52 = 3640$$

$$\equiv 41 \pmod{61}$$

$$3640 \div 61 = 59.67 \dots$$

$$3640 = 61 \times 59 + 41$$

$$\textcircled{2} \quad 13n \equiv 4 \pmod{8}, \quad (0 \leq n \leq 7)$$

$$13 \Rightarrow 8 \times 1 + 5$$

$$5n \equiv 4 \pmod{8}$$

$$\text{inverse} \Rightarrow (5 \pmod{8})$$

$$\begin{array}{l|l} \text{gcd}(5, 8) \Rightarrow & \underline{\underline{5}} \\ & -1 \\ & 1+1=2 \\ & -1-2=-3. \end{array} \quad \swarrow \text{inverse}$$

$$5x - 3 \equiv 1 \pmod{8}$$

$$\left| \begin{array}{l} -3+8 = +5 \end{array} \right.$$

$$\Rightarrow 5 \pmod{8}$$

$$\text{back to } 5n \equiv 4 \pmod{8} \Rightarrow$$

$$\times 5 \Rightarrow n \equiv 5 \times 4 \pmod{8}$$

$$n \equiv 20 \pmod{8}$$

$$n = 4$$

let's verify $\Rightarrow$

$$13n \equiv 4 \pmod{8}$$

$$13 \times 4 = 52 \equiv 4 \pmod{8}$$